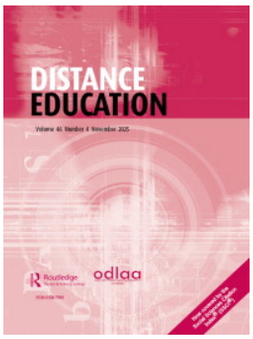

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ARTICLE

Empowering teacher training with AI: factors shaping the development and use of open educational resources

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ABSTRACT
The integration of artificial intelligence (AI) into open educational resource (OER) development presents new opportunities for enhancing educational experiences. However, the factors shaping pre-service teachers' adoption of AI-assisted OER development still unexplored. This study examines the key determinants of pre-service teachers' (psts) behavioral intentions to develop and use AI-assisted OER, drawing from Uses and Gratifications Theory and related constructs. Data were collected from 240 psts. Participants received 10 weeks training on AI-supported OER development. Data analysis was conducted performing structural equation modeling and bootstrapping. The findings reveal that compatibility, enjoyment, task efficiency, and information seeking significantly influence both trust and behavioral intention. However, trust did not. Social factors yielded mixed results. While social influence positively affected behavioral intention, it did not significantly impact compatibility or trust. Similarly, social interaction fostered trust but did not directly influence behavioral intention. While peer and instructor opinions encourage adoption, they do not necessarily enhance AI's trustworthiness or suitability for educational tasks. The study contributes to the theoretical understanding of AI acceptance in education by refining existing adoption models with social and cognitive factors. Findings suggest that training programs should emphasize AI's alignment with teaching tasks and focus on enhancing user enjoyment to drive adoption.

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KEYWORDS
Artificial intelligence; open educational resources; pre-service teachers; uses and gratifications theory; structural equation modeling

Introduction

As digital educational environments and tools become increasingly prevalent, open educational resources (OER) are gaining prominence in education (Otto, 2019). One of the key priorities in global education policy is promoting participation in OER initiatives. According to UNESCO, interest in OER continues to grow (Smirani & Boulahia, 2022). Recognized as a key element in enhancing educational quality and promoting equitable access, OER supports flexible engagement with digital content by fostering an open sharing culture (Laufer et al., 2021). OER promotes greater flexibility and customization

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in learning by enabling individuals to retain, reuse, modify, adapt, and share educational content (Wiley et al., 2014). As freely accessible and adaptable materials, OER empower educators and students to create, modify, and share resources, enriching the learning experience (Li et al., 2024; Mncube et al., 2024). By expanding access to high-quality resources, OER fosters a more inclusive and equitable education system, ensuring that diverse learners can benefit from freely available and adaptable materials (Hilton, 2020).

Although OER offers significant potential through its unique features and opportunities, AI has the capacity to elevate it even further. In the context of OER production and use, AI can contribute valuable enhancements such as personalization, adaptability, increased creativity, innovation, and greater efficiency, making educational resources more dynamic and impactful. AI-assisted tools can also automate content creation, allowing educators to generate high-quality materials in less time, while tailoring resources to diverse learner needs (Bozkurt, 2023; Khosravi et al., 2023; Morales-Chan et al., 2024). As the integration of AI in OER production and use evolves, research in this area is crucial to understand its full potential, identify best practices, and influential factors. Studies exploring these dimensions will contribute to the development of more effective, inclusive, and sustainable educational resources in the digital age.

A culture of sharing educational resources has become a critical necessity in modern education (Adil et al., 2024). Andrade et al. (2011) identified critical challenges hindering OER use which are significant gaps in educators' skills and time for OER creation or adaptation as major obstacles, while limited availability of resources in users' native languages emerged as persistent structural barriers. Additionally, more recent research has further emphasized these multifaceted challenges such as low resource quality, limited awareness, lack of time among OER creators, and technical skill gaps (Adil et al., 2024). Despite the growing development and positive impact of OER in higher education, its adoption remains largely dependent on individual educators (Jung & Lee, 2020). However, key aspects, such as educators' perceptions and adoption of OER, have not been fully explored. While existing research highlights technological challenges and barriers to OER adoption, its widespread use remains limited (Otto, 2019). Integrating open educational practices (OEP) into teacher training through contextualized, hands-on experiences can empower educators to create, adapt, and publish OER, and this can ultimately help address the shortage of high-quality, classroom-ready open materials (Tang & Bao, 2021). Although various obstacles continue to hinder OER integration, and studies examining the factors influencing OER adoption and utilization remain limited (Alkawsi et al., 2025; Menzli et al., 2022). Moreover, when the few available studies on OER adoption and use are examined through the lens of AI integration, relevant research becomes virtually nonexistent. This highlights a clear gap in the literature and underscores the need to identify the motivational factors shaping educators' willingness to develop OER with AI assistance. In particular, understanding the key drivers behind pre-service teachers' acceptance and adoption of AI-supported OER creation is essential for fostering broader engagement and sustainable implementation.

The purpose of this study is to examine the factors influencing pre-service teachers' tendency to develop and use OER with the help of artificial intelligence. Specifically, the study aims to explore how motivational and technological factors shape their behavioral intention, trust, and perceived compatibility regarding AI-assisted OER

creation. To achieve this, a model is proposed by utilizing the Uses and Gratifications Theory (UGT) (information seeking, task efficiency, social interaction, and enjoyment) and additional motivational variables. Additionally, this study adopts only behavioral intention (BI) as an outcome variable, reflecting the extent to which pre-service teachers intend to engage in AI-assisted OER development. Since the successful adoption of AI tools in OER depends on user willingness, BI is a crucial determinant in predicting actual engagement with this technology. This approach provides a comprehensive framework for understanding both the technological and psychological dimensions of AI-supported OER adoption.

Significance of the study

This study makes a significant contribution to the literature by utilizing the Uses and Gratifications Theory (UGT) and motivational external variables to examine pre-service teachers' tendency to develop and use open educational resources produced with the help of artificial intelligence. By integrating TAM's behavioral intention construct (Davis, 1989) with UGT's emphasis on motivations and gratifications (Katz et al., 1973), along with trust and compatibility as additional outcome variables, this study provides a holistic understanding of AI-assisted OER adoption. This combined framework allows for the examination of both external technological influences and internal motivational drivers (Şahin et al., 2025), offering a comprehensive perspective on how to encourage the development and use of AI-assisted OER in teacher education. UGT is particularly relevant to this study because the adoption of AI-assisted OER is not solely a rational decision based on ease of use and usefulness but also a motivational choice driven by personal and professional needs. Understanding these motivations provides deeper insights into why educators may or may not trust AI-generated content and how they perceive its compatibility with their teaching practices.

By focusing on a participant group of future educators who received structured training on developing and utilizing AI-assisted OER, this study offers valuable empirical insights into how pedagogical and technological factors influence their adoption behavior. Furthermore, this research fills a critical gap in the literature by being the first to explore the development and use of AI-assisted OER from the perspective of UGT, highlighting the role of user motivations in adopting AI tools for educational content creation. As AI technologies increasingly support the production of open educational resources, understanding pre-service teachers' perceptions and motivations is essential for designing effective professional development programs and policy interventions. The study's findings can guide educators, policymakers, and instructional designers in fostering AI literacy and promoting sustainable practices in OER creation, ultimately contributing to more accessible, diverse, and high-quality educational materials in the digital age.

Theoretical background

Outcome variables

This study examines behavioral intention, compatibility, and trust as key outcome variables to understand the factors influencing pre-service teachers' adoption of

AI-assisted open educational resources. These constructs help explain both the technological and psychological aspects of the adoption process. Behavioral intention refers to an individual's willingness and commitment to engage in a specific behavior (Davis et al., 1989)-in this case, developing and using AI-assisted OER. It is a strong predictor of actual adoption (Venkatesh et al., 2003), as individuals with higher behavioral intention are more likely to incorporate AI tools into their OER practices. Since AI in education is still an emerging technology, identifying the factors that enhance or hinder behavioral intention is crucial for its successful integration.

Trust refers to the extent to which individuals perceive AI-assisted OER as reliable, accurate, and ethically sound. Given that AI-generated content can raise concerns about credibility, bias, and control, trust becomes a key determinant of whether educators feel confident using such tools (Baek & Kim, 2023). When trust in AI is high, educators are more likely to adopt a positive attitude toward its use, which in turn strengthens their behavioral intention. Research on educational technology acceptance consistently shows that trust in AI systems-particularly in their fairness, reliability, and accuracy—significantly predicts both students' and educators' willingness to integrate AI into educational processes (Chai et al., 2024; Khosravi et al., 2022). Furthermore, in academic environments where AI tools are perceived as transparent, dependable, and ethically designed, the likelihood of their adoption increases (Akgun & Greenhow, 2022; Rana et al., 2024). This underscores the pivotal role of trust in fostering AI adoption, shaping user perceptions, and promoting meaningful engagement with AI-assisted OER.

Compatibility may serve as a critical factor in technology adoption, referring to the extent to which a new technology aligns with users' existing values, needs, and work practices (Venkatesh & Davis, 2000). If educators perceive AI-generated OER as compatible with their workflow and pedagogical beliefs, they are more likely to integrate these resources into their teaching. High compatibility reduces resistance to new technologies and increases both trust and behavioral intention. On the other hand, if AI-generated resources require excessive adaptation or conflict with their teaching philosophy, they may resist adoption despite recognizing its potential benefits. In the context of trust, if AI seamlessly integrates into their existing practices without requiring significant adjustments, their confidence in its usability and effectiveness may increase. Conversely, if AI-generated OER feels misaligned with their pedagogical approaches, skepticism and distrust may arise.

Specifically, behavioral intention represents a key indicator of the likelihood that pre-service teachers will adopt AI-assisted OER. Compatibility captures the perceived alignment between AI-assisted OER development and pre-service teachers' pedagogical values, instructional needs, and teaching contexts. Aligning AI solutions with existing practices and values facilitates their adoption and integration, as compatibility has been identified as a key enabler of generative AI adoption (Abulail et al., 2025; Raman et al., 2024). In educational contexts, users who perceive generative AI as compatible with their practices and needs are more likely to find it easy to use and beneficial for achieving their goals (Almogren et al., 2024). Trust, in turn, reflects users' confidence in AI's reliability, transparency, and pedagogical appropriateness. Human–AI trust is as crucial as the AI's technical capabilities (Henrique & Santos, 2024) and serves as a primary mechanism driving the acceptance and adoption of AI service

providers and generative AI (Choudhury & Shamszare, 2023; Jiang et al., 2024). It is shaped by factors related to the human, the AI system, and the context, yet few empirical studies have examined these influences (Kaplan et al., 2023), and the specific application or use case can further determine trust and acceptance (Gillespie et al., 2023). Importantly, in the context of AI-assisted OER development, perceived compatibility can strengthen trust, and together, these factors jointly enhance behavioral intention. This combination is especially powerful for investigating AI adoption because it integrates both cognitive evaluations (compatibility) and affective judgments (trust), providing a comprehensive understanding of pre-service teachers' willingness to develop and use AI-assisted OER. From this point of view, the following hypotheses were formulated:

H1. Trust has a significant influence on behavioral intention.

H2. Compatibility has a significant influence on (a) trust and (b) behavioral intention.

Uses and gratifications theory

Uses and Gratifications Theory (UGT) is a psychological and communication framework that examines why individuals actively seek out specific media and technologies to fulfill their needs (Katz et al., 1973; Kim et al., 2021). According to UGT, users engage with technology for various gratifications, including information-seeking, social interaction, entertainment, and productivity enhancement (Şahin et al., 2025). This theory emphasizes the active role of users in selecting and utilizing technologies based on their personal motivations and anticipated benefits. Grounded in UGT, this study investigates how key motivational factors—information seeking, task efficiency, social interaction, and enjoyment—shape educators' perceptions of compatibility, trust, and behavioral intention in adopting AI-assisted open educational resources (OER). As AI becomes an increasingly prominent tool in education, understanding the psychological drivers behind educators' adoption of AI-assisted OER is critical. In this context, UGT provides a robust framework for examining how educators perceive AI's alignment with their professional practices (compatibility), its reliability and credibility (trust), and their willingness to integrate it into teaching (behavioral intention). Building on the existing literature, this study conceptually and empirically integrates these motivational dimensions within the context of AI-assisted OER adoption (Şahin et al., 2025; Zimmermann et al., 2023).

Educators frequently engage in information seeking to enhance their teaching strategies, improve content quality, and stay updated with emerging educational technologies. AI-assisted OER can significantly contribute to this process by providing quick access to diverse, high-quality educational materials. When educators perceive AI-generated content as accurate, relevant, and credible, they are more likely to see AI-assisted OER as compatible with their teaching practices. Additionally, trust in AI systems grows when users recognize AI's ability to provide reliable and bias-free information. As a result, educators who actively seek information are more inclined to develop a behavioral intention to engage with AI-assisted OER, as it effectively meets their need for accessible and high-quality educational resources. Task efficiency plays a crucial role in the adoption of AI-assisted OER, as educators increasingly seek

tools that enhance productivity and streamline workload management. AI-powered tools can automate content creation, personalize learning materials, and optimize instructional design, making them highly compatible with educators' existing workflows. The perception that AI improves task efficiency strengthens behavioral intention, as educators recognize its potential to reduce workload while enhancing teaching effectiveness. When technology aligns with professional efficiency goals, its adoption becomes a natural extension of educators' daily practices.

Enjoyment is another significant factor influencing technology adoption, as positive experiences with AI-assisted OER enhance engagement, motivation, and ease of use. If educators find the process of creating AI-assisted OER enjoyable, they are more likely to perceive it as compatible with their teaching style and professional needs. Moreover, enjoyment fosters trust, as users who have satisfying and seamless experiences with AI tools tend to view them as reliable, user-friendly, and beneficial. Ultimately, when educators derive intrinsic motivation and satisfaction from using AI, their behavioral intention to integrate AI-assisted OER strengthens. Social interaction also plays a critical role in AI-assisted OER adoption. AI tools not only facilitate peer collaboration but also allow educators to actively interact with AI itself, enhancing the co-creation and sharing of educational content. The ability of AI to generate, refine, and adapt learning materials fosters meaningful engagement and dynamic collaboration, making AI-assisted OER an attractive tool for educators. This interactive nature further strengthens behavioral intention, as educators become more inclined to use AI in their teaching practices. Additionally, when AI tools support effective communication and collaboration, trust in their capabilities increases, leading to greater adoption. In this regard, the following hypotheses were proposed:

H3. Information seeking has a significant influence on (a) compatibility, (b) trust, and (c) behavioral intention.
H4. Task efficiency has a significant influence on (a) compatibility and (b) behavioral intention.
H5. Enjoyment has a significant influence on (a) compatibility, (b) trust, and (c) behavioral intention.
H6. Social interaction has a significant influence on (a) trust and (b) behavioral intention.

Social influence

Social influence, defined as the degree to which individuals perceive that important others (e.g., colleagues, administrators, professional communities) encourage or expect them to adopt a technology (Ajzen & Fishbein, 1980; Venkatesh & Morris, 2000), plays a significant role in shaping educators' attitudes toward AI-assisted OER. When educators observe that their peers trust AI-generated content and view it as a credible and reliable tool, their own trust in AI systems is likely to strengthen.

Positive recommendations, shared experiences, and institutional endorsements can alleviate concerns about AI-generated content's accuracy, bias, and ethical considerations, fostering greater confidence in its use. Furthermore, social influence affects compatibility, as educators who see AI tools being successfully integrated into teaching practices within their professional networks are more likely to perceive them as aligned with their pedagogical needs and instructional workflows. This alignment increases

the likelihood of adoption, as educators recognize AI-assisted OER as a valuable and practical resource.

Social influence has a potential to drive tendency by reinforcing the perceived benefits and acceptability of AI within educational environments (Foroughi et al., 2024). Ultimately, social influence shapes behavioral intention, as the encouragement and endorsement of AI tools by respected peers and institutions enhance educators' willingness to experiment with and adopt AI-assisted OER. In professional environments where AI is positioned as an effective and beneficial tool, educators are more motivated to integrate it into their teaching, reinforcing the importance of social influence in technology adoption. Previous studies on AI also present supporting results (Al-Emran et al., 2024; Sing et al., 2022). Consequently, the following hypothesis was created:

H7. Social influence has a significant influence on (a) compatibility, (b) trust, and (c) behavioral intention.

The proposed model of the study is presented in Figure 1.

Methodology

Participants

The study was conducted with 240 senior pre-service teachers enrolled in the Open and Distance Learning course, representing eight different teacher education departments. As future educators, pre-service teachers will be expected to design, adapt, and use OERs in their future professional practice, including in potential ODL environments, especially for blended learning environments and for their students coming from lower socio-economic backgrounds. The Open and Distance Learning course that the participants of this study took aimed to provide teacher candidates with a comprehensive

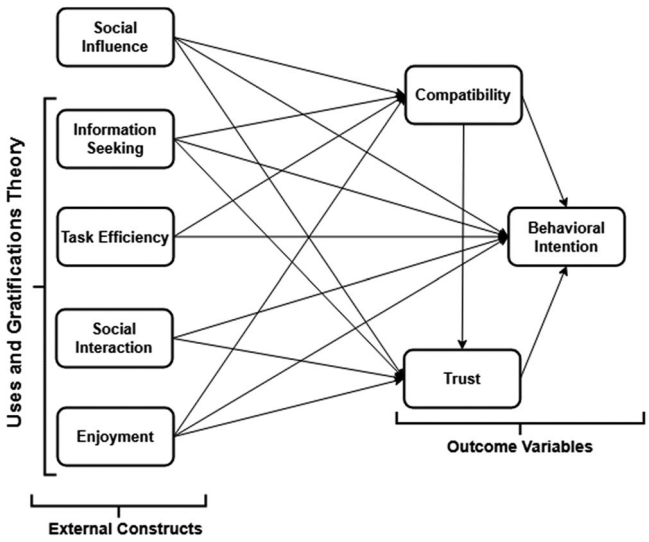

Figure 1. The proposed model of the study.

Table 1. Demographic characteristics of the pre-service teachers.

Pre-service Teachers		*f*	%
Department	Classroom Teaching	45	18.75
	English Language Teaching	36	15.00
	Early Childhood Education	33	13.75
	Mathematics Teaching	32	13.33
	Science Education	25	10.42
	Psychological Counseling and Guidance	24	10.00
	Social Studies Education	23	9.58
	Art & Music Education	22	9.17
Gender	Female	151	62.9
	Male	89	37.1
Age	Min	Max	M (SD)
	20	25	23.56 (3.01)

understanding of the principles and practices of open and distance education, including distance education systems, open educational resources (OERs), and open educational practices (OEPs). The use of AI-assisted OER creation was introduced as a practical and pedagogically relevant activity aligned with the values of openness, accessibility, and flexibility that underpin distance education. Over a 10-week period, participants received training on developing and utilizing open educational resources with the assistance of AI tools such as ChatGPT, CoPilot, and Deepseek. Data were collected through an online survey, ensuring accessibility and ease of participation. Prior to data collection, the study was approved by the institutional review board. Participants were fully informed about the study's purpose, and it was explicitly stated that their involvement was completely voluntary. The sample comprised pre-service teachers from eight teacher education departments: Classroom Teaching (n = 45, 18.75%), English Language Teaching (n = 36, 15%), Early Childhood Education (n = 33,13.75%), Mathematics Teaching (n = 32, 13.33%), Science Education (n = 25, 10.42%), Psychological Counseling and Guidance (n = 24, 10%), Social Studies Education (n = 23, 9.58%), and Art & Music Education (n = 22, 9.17%). With a more diverse and representative sample, the study enhanced the generalizability and external validity of the results. Demographic characteristics of the pre-service teachers are summarized in Table 1.

Data collection tool

The study's data collection tool consisted of two sections. The first section gathered demographic and personal information, while the second section contained 29 scale items measuring eight dimensions on a 5-point Likert scale (1: Completely Disagree – 5: Completely Agree). The scale items for enjoyment, compatibility, social influence, and behavioral intention were adapted from Şahin et al. (2021, 2025), while the items for information seeking, task efficiency, social interaction, and trust were adapted from Baek and Kim (2023). To ensure content and face validity, the measurement tool was first reviewed by field experts, who provided feedback on item clarity and relevance. A pilot study was then conducted with a group of five pre-service teachers, assessing the comprehensibility of the items and estimating the average completion time. Based on the feedback from these validation steps, necessary revisions and refinements were made before proceeding with the main data collection phase.

Data analysis

The data were analyzed using SPSS 27 and SmartPLS 4. In SPSS, descriptive statistics were calculated, and Mahalanobis distance was used to identify extreme values. In SmartPLS, the measurement and structural model were evaluated by performing factor analysis, structural equation modeling (SEM) and bootstrapping analysis. To ensure construct validity, item loadings, Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE) were examined for convergent validity.

Convergent validity assesses whether the items intended to measure the same construct are highly correlated, ensuring that the construct is consistently represented across its indicators. A construct demonstrates strong convergent validity when item loadings exceed 0.70, composite reliability (CR) is above 0.70, and AVE values are greater than 0.50 (Hair et al., 2017; 2021). For discriminant validity, the Fornell-Larcker criterion and Heterotrait-Monotrait (HTMT) ratio were assessed.

Discriminant validity ensures that a construct is empirically distinct from other constructs in the model, meaning it captures a unique concept. The Fornell-Larcker criterion requires that a construct's AVE is greater than its squared correlations with other constructs, while the HTMT ratio should remain below 0.90 to confirm that constructs are not excessively similar (Fornell & Larcker, 1981; Henseler et al., 2015). Additionally, variance inflation factor (VIF) values were analyzed to check for multicollinearity among predictive constructs. Finally, model fit was evaluated using the standardized root mean square residual (SRMR).

Results

Measurement model evaluation

The measurement model was assessed to ensure reliability, validity, and overall adequacy before proceeding with the structural model evaluation. Convergent validity was examined through Cronbach's alpha, Composite Reliability (CR), Average Variance Extracted (AVE), and factor loadings (see Table 2). All constructs demonstrated Cronbach's alpha values above 0.70 and CR values exceeding 0.70, indicating strong internal consistency and reliability (Hair et al., 2021). These results confirm that the items within each construct consistently measure the intended latent variables. Additionally, all factor loadings exceeded the recommended threshold, confirming that each indicator contributes meaningfully to its respective construct. The AVE values were above 0.50 for all constructs, demonstrating that the majority of the variance in the observed variables is explained by their respective latent constructs (Hair et al., 2017).

To ensure discriminant validity, the Fornell-Larcker criterion and Heterotrait-Monotrait (HTMT) ratio were tested (see Table 3 & Table 4). According to the Fornell-Larcker criterion, the square root of each construct's AVE exceeded its correlations with other constructs, confirming that each construct is more strongly related to its own indicators than to other constructs in the model (Fornell & Larcker, 1981). Furthermore, HTMT ratios remained below the 0.90 threshold, providing additional evidence of sufficient discriminant validity (Henseler et al., 2015).

Table 2. Convergent validity.

Factors	Item	Item Loading	α	CR	Ave
Behavioral Intention	BI1	0.882	0.844	0.846	0.762
	BI2	0.876			
	BI3	0.861			
Compatibility	CMP1	0.678	0.810	0.829	0.640
	CMP2	0.828			
	CMP3	0.873			
	CMP4	0.807			
Trust	TRST1	0.844	0.759	0.783	0.676
	TRST2	0.883			
	TRST3	0.733			
Task Efficiency	TE1	0.770	0.778	0.780	0.600
	TE2	0.766			
	TE3	0.762			
	TE4	0.800			
Information Seeking	IS1	0.872	0.835	0.839	0.752
	IS2	0.895			
	IS3	0.834			
Enjoyment	ENJ1	0.770	0.882	0.894	0.740
	ENJ2	0.875			
	ENJ3	0.897			
	ENJ4	0.892			
Social Interaction	SINT1	0.743	0.811	0.820	0.638
	SINT2	0.779			
	SINT3	0.853			
	SINT4	0.817			
Social Influence	SI1	0.726	0.752	0.761	0.569
	SI2	0.691			
	SI3	0.788			
	SI4	0.807			

Multicollinearity was evaluated using Variance Inflation Factor (VIF) values, all of which remained below 5, indicating the absence of multicollinearity issues (Hair et al., 2021). Additionally, the Standardized Root Mean Square Residual (SRMR) value was 0.059, which falls below the 0.08 threshold, indicating a good model fit (Hu & Bentler, 1999). Overall, these findings confirm that the measurement model meets all established reliability and validity criteria, ensuring that the constructs are well-defined and appropriate for further structural model analysis.

Structural model evaluation

Following the assessment of the measurement model, the structural model was evaluated to examine the hypothesized relationships between constructs. The analysis was conducted using Partial Least Squares Structural Equation Modeling (PLS-SEM) and bootstrapping in SmartPLS 4, and the results were assessed based on path coefficients, t-values, significance levels (Table 5), and explained variance (R^2) (Figure 2).

The findings indicate that trust did not significantly influence behavioral intention (H1 not supported), suggesting that while trust is an important consideration in AI-assisted OER adoption, it may not directly drive pre-service teachers' intention to use these technologies. However, compatibility significantly influenced both trust (H2a supported) and behavioral intention (H2b supported), highlighting that when pre-service teachers perceive AI-assisted OER as compatible with their teaching practices, they are more likely to trust the technology and intend to use it. Regarding information-seeking behavior, the results show that it significantly influenced compatibility (H3a supported) and behavioral intention (H3c supported), indicating that pre-service teachers who actively seek information are more likely to perceive AI-assisted OER as compatible with their needs and develop an intention to adopt them. However, information-seeking did not significantly impact trust (H3b not

Table 3. Fornell-Larcker criterion (discriminant validity).

Factors	BI	Cmp	Enj	Is	SI	Sint	TE	Trst
BI	0.873							
CMP	0.639	0.800						
ENJ	0.682	0.669	0.860					
IS	0.625	0.589	0.588	0.867				
SI	0.565	0.488	0.601	0.403	0.754			
SINT	0.469	0.458	0.547	0.479	0.383	0.799		
TE	0.576	0.605	0.613	0.626	0.465	0.503	0.775	
TRST	0.468	0.553	0.540	0.502	0.422	0.468	0.423	0.822

Note. Numbers in bold represent the square root of the average variance extracted (AVE).

Table 4. HTMT ratio matrix (discriminant validity).

Factors	BI	Cmp	Enj	Is	SI	Sint	TE	Trst
BI								
CMP	0.765							
ENJ	0.780	0.774						
IS	0.744	0.714	0.689					
SI	0.682	0.596	0.701	0.484				
SINT	0.564	0.550	0.653	0.587	0.473			
TE	0.706	0.749	0.736	0.775	0.577	0.631		
TRST	0.579	0.713	0.658	0.627	0.526	0.581	0.531	

Table 5. Structural model.

	Path	CoEff.	t-Value	p-Value	Vif	Results
H1	TRST → BI	−0.018	0.254	0.799	1.695	Not Supported
H2a	CMP → TRST	0.249	3.379	0.001	2.069	**Supported**
H2b	CMP → BI	0.197	2.874	0.004	2.279	**Supported**
H3a	IS → CMP	0.207	3.218	0.001	1.846	**Supported**
H3b	IS → TRST	0.160	1.874	0.061	1.778	Not Supported
H3c	IS → BI	0.245	3.911	0.000	2.031	**Supported**
H4a	TE → CMP	0.210	3.277	0.001	1.970	**Supported**
H4b	TE → BI	0.051	0.814	0.416	2.130	Not Supported
H5a	ENJ → CMP	0.368	5.492	0.000	2.202	**Supported**
H5b	ENJ → TRST	0.131	1.588	0.112	2.569	Not Supported
H5c	ENJ → BI	0.252	3.780	0.000	2.663	**Supported**
H6a	SINT → TRST	0.171	2.698	0.007	1.522	**Supported**
H6b	SINT → BI	0.034	0.538	0.591	1.618	Not Supported
H7a	SI → CMP	0.086	1.537	0.124	1.604	Not Supported
H7b	SI → TRST	0.092	1.282	0.200	1.606	Not Supported
H7c	SI → BI	0.188	3.050	0.002	1.639	**Supported**

supported), suggesting that trust in AI may be shaped by factors beyond just information availability.

Task efficiency was found to significantly enhance compatibility (H4a supported), demonstrating that when pre-service teachers perceive AI tools as efficient for their tasks, they view them as more aligned with their professional needs. However, task efficiency did not directly influence behavioral intention (H4b not supported), implying that efficiency alone may not be sufficient to drive adoption without additional motivational factors. Similarly, enjoyment played a significant role in compatibility (H5a supported) and behavioral intention (H5c supported), reinforcing the idea that when pre-service teachers find AI-assisted OER enjoyable to use, they are more likely to perceive them as fitting their teaching practices and develop an intention to integrate them. However, enjoyment did not significantly affect trust (H5b not supported),

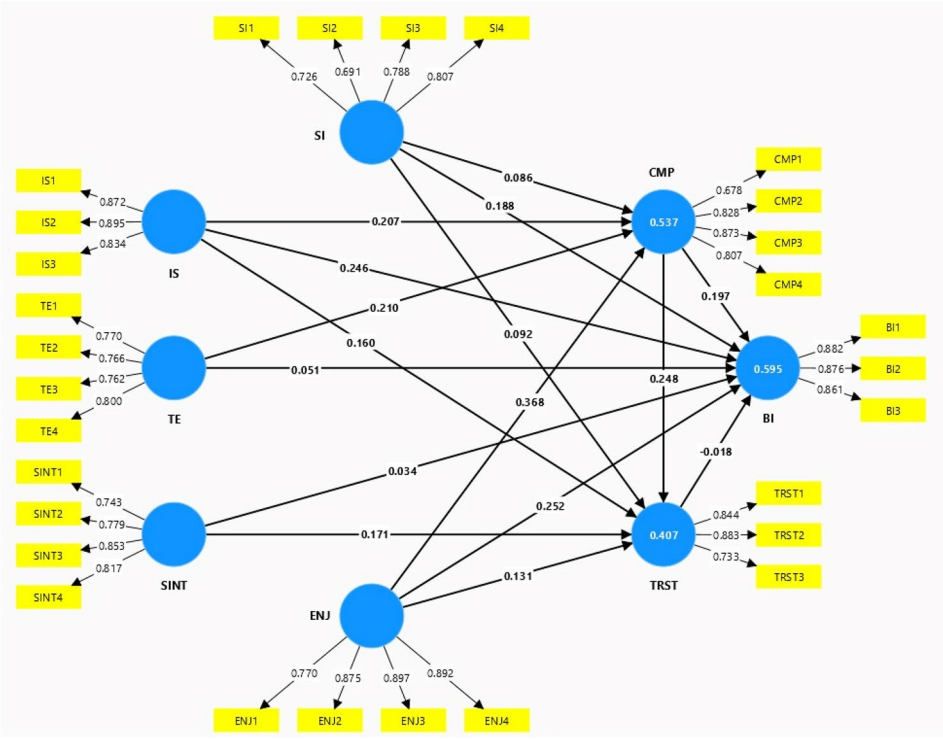

Figure 2. Partial least squares - structural equation modeling results.

suggesting that while enjoyment enhances adoption, it does not necessarily increase confidence in AI's reliability or fairness.

Social interaction significantly influenced trust (H6a supported) but did not directly affect behavioral intention (H6b not supported). This finding implies that collaborative and interactive engagement with AI-assisted OER fosters trust but does not necessarily lead to adoption unless accompanied by other motivating factors. Lastly, social influence did not significantly affect compatibility (H7a not supported) or trust (H7b not supported), indicating that external social pressures alone do not necessarily make AI-assisted OER feel more aligned with pre-service teachers' teaching practices or more trustworthy. However, social influence had a significant positive impact on behavioral intention (H7c supported), suggesting that peer recommendations, institutional encouragement, or broader societal trends can drive pre-service teachers toward AI-assisted OER adoption.

Discussion

The purpose of this study was to examine the factors influencing pre-service teachers' tendency to develop and use OER with the assistance of artificial intelligence. Specifically, the study aimed to explore how motivational and social factors shape their behavioral intention, trust, and perceived compatibility regarding AI-assisted OER creation. To achieve this, a model was proposed by utilizing the Uses and Gratifications Theory (UGT) (information seeking, task efficiency, social interaction, and enjoyment

dimensions) and additional variables. While many findings aligned with previous research, unsupported hypotheses revealed meaningful differences related to pre-service teachers' developmental stage and the emerging nature of AI-assisted OER creation.

The findings supported the hypotheses Compatibility → Trust and Compatibility → Behavioral Intention, indicating that when AI-assisted OER development aligns with pre-service teachers' teaching activities and expectations, it positively influences both their trust in AI-assisted OER and their intention to use it. However, the Trust → Behavioral Intention hypothesis was not supported, suggesting that trust in AI does not directly translate into a stronger intention to adopt AI-assisted OER. Although compatibility has been very much underexplored in the context of AI acceptance, the findings of this study are consistent with previous research (Jafari et al., 2024), highlighting its significant role in shaping users' perceptions and adoption decisions. In contrast, the rejection of the Trust → Behavioral Intention hypothesis diverges from prior studies (e.g., Ali et al., 2023; Baek & Kim, 2023; Choudhury & Shamszare, 2023; Emon et al., 2023), which have generally identified trust as a crucial factor in technology acceptance. Perceived trust, however, can be influenced by situational factors such as task difficulty, perceived performance, and task success or failure (Bitkina et al., 2020), which may have affected how pre-service teachers evaluated AI-assisted OER in this study. A possible explanation for this finding is that while trust in AI is important, it may not be the primary driver of behavioral intention in this context. Pre-service teachers might prioritize practical benefits, such as efficiency or pedagogical alignment, over their level of trust in AI systems when deciding whether to adopt these tools. Additionally, trust alone may not be sufficient to drive behavioral intention without additional motivational reinforcements, such as institutional support, peer influence, or hands-on experience demonstrating AI's effectiveness in real teaching scenarios. This interpretation aligns with the idea that pre-service teachers, who are still developing professional confidence and pedagogical identity, may hesitate to act on trust alone without seeing concrete instructional value. Their trust may therefore remain a passive belief rather than an active motivator of adoption. This suggests that fostering perceived usefulness and compatibility may be more critical than trust in encouraging AI-assisted OER adoption.

The analysis results showed that the Information Seeking → Compatibility and Information Seeking → Behavioral Intention hypotheses were supported, suggesting that pre-service teachers' motivation to seek information plays a crucial role in their adoption of AI-assisted OER. Specifically, those who actively search for information are more likely to perceive AI-assisted OER as compatible with their teaching practices and are more inclined to use it. However, the rejection of the Information Seeking → Trust hypothesis suggests that information-seeking behaviors do not significantly influence pre-service teachers' trust in AI for OER development. While no previous studies have specifically examined the causal relationship between Information Seeking → Compatibility in the context of OER, some AI and technology acceptance studies have provided partial support for the Information Seeking → Behavioral Intention and Information Seeking → Trust relationships (Baek & Kim, 2023; Deng et al., 2015). A possible explanation for this unsupported relationship is that while seeking

information enhances familiarity with AI tools, it does not necessarily address concerns related to reliability, bias, or ethical considerations. In the context of pre-service teacher education, this may stem from limited evaluative experience. While they can gather information, they may lack the professional lens to assess AI's credibility or pedagogical appropriateness. Consequently, their trust in AI is more likely shaped by instructor guidance, institutional endorsement, or observed instructional applications than by independent information-seeking efforts. Trust in AI may be shaped more by direct experiences, institutional endorsements, or perceived transparency rather than the quantity of information obtained.

The results revealed that the Task Efficiency → Compatibility hypothesis was supported, while the Task Efficiency → Behavioral Intention hypothesis was not. This suggests that pre-service teachers perceive AI assistance in OER development as beneficial for streamlining tasks and aligning with their workflow expectations. However, this perceived efficiency does not directly translate into a stronger intention to develop and use AI-assisted OER. Although no prior study has explicitly examined the Task Efficiency → Compatibility relationship, previous research supports the notion that task efficiency is a key factor in technology adoption (e.g., Al-Mamary et al., 2024; Baek & Kim, 2023). However, these findings contrast with the rejected Task Efficiency → Behavioral Intention hypothesis in this study. A possible explanation for this finding is that while AI can enhance productivity, other factors, such as pedagogical value, perceived ease of use, or institutional support, may play a more significant role in shaping behavioral intention. This may be particularly true for pre-service teachers, who face fewer time constraints than in-service educators and are more concerned with learning to design meaningful learning materials than with task speed. For them, efficiency may be appreciated but not decisive unless it directly supports instructional quality or creativity in OER development. Additionally, pre-service teachers might view AI as a useful tool for improving efficiency but remain hesitant to fully integrate it into their teaching due to concerns about content quality, ethical considerations, or a lack of familiarity with AI-driven OER creation.

The Enjoyment → Compatibility and Enjoyment → Behavioral Intention hypotheses were supported, while Enjoyment → Trust hypothesis was rejected. These findings suggest that when pre-service teachers perceive AI-assisted OER development as enjoyable, they are more likely to see it as aligned with their expectations and professional needs, thereby increasing their intention to use it. However, their sense of enjoyment does not necessarily translate into greater trust in AI. These results are consistent with prior research indicating that enjoyment enhances user satisfaction and strengthens adoption tendencies in AI-driven tools (Kim et al., 2025; Kim & Baek, 2025). The rejection of Enjoyment → Trust may indicate that pre-service teachers may appreciate the creative and engaging aspects of AI but remain cautious due to ethical concerns related to data privacy, authorship, and the potential for AI-generated misinformation. Furthermore, pre-service teachers might feel comfortable and entertained while using AI but still hesitate to fully rely on it for critical educational tasks. For pre-service teachers, enjoyment may function as a short-term motivator driven by curiosity and novelty rather than a sign of enduring confidence. Their enjoyment may coexist with uncertainty, as they lack extensive experience validating AI outputs for pedagogical soundness, which may limit the development of genuine trust.

The results indicated that the Social Interaction → Trust hypothesis was supported, whereas the Social Interaction → Behavioral Intention hypothesis was not. This suggests that AI's ability to facilitate interactions enhances trust in the tool but does not directly influence pre-service teachers' intention to adopt it. While previous research presents conflicting findings regarding the Social Interaction → Trust relationship (Baek & Kim, 2023), the positive association in this study implies that AI-driven collaboration features may strengthen users' confidence in the technology. However, the lack of support for the Social Interaction → Behavioral Intention relationship suggests that while AI can foster engagement and interaction, these factors alone may not be strong enough to drive adoption decisions. To date, no prior studies have specifically investigated the direct link between social interaction and behavioral intention in the context of AI-supported OER adoption, highlighting a need for further research in this area. This may reflect the developmental context of pre-service teachers, who are still forming professional communities and may not yet rely on peer networks to guide technology decisions. Their adoption intentions may depend more on perceived pedagogical benefits than on social connectedness or collaborative experiences.

The analysis results showed that while the Social Influence → Behavioral Intention hypothesis was supported, the Social Influence → Compatibility and Social Influence → Trust hypotheses were not. These findings suggest that the opinions of influential individuals, such as mentors, instructors, or colleagues, play a significant role in shaping pre-service teachers' intention to adopt AI-supported OER. However, the perceived compatibility of AI-assisted OER for professional practice and the level of trust in the technology appear to be unaffected by social influence, indicating that personal assessments of compatibility and trust are shaped more by direct experience and individual perceptions rather than external opinions. Although no prior studies have explicitly examined the rejected Social Influence → Compatibility and Social Influence → Trust relationships, existing literature presents mixed findings on the Social Influence → Behavioral Intention relationship (e.g., Emon et al., 2023; Foroughi et al., 2024; Parveen et al., 2024; Strzelecki, 2024). These discrepancies highlight the need for further research to better understand the role of social influence on AI-supported OER adoption, particularly regarding its impact on trust and compatibility. In the teacher education context, these findings may stem from the limited institutional modeling of AI-assisted OER use. Pre-service teachers may not yet have enough exposure to trusted mentors or exemplars who actively integrate AI into instructional design, making their evaluations more self-driven and exploratory.

Conclusion and implications

This study explored the factors influencing pre-service teachers' adoption of AI-assisted open educational resources (OER) by integrating the Uses and Gratifications Theory (UGT) with key adoption constructs. The findings provide valuable insights into how motivations such as information seeking, task efficiency, enjoyment, and social interaction shape perceptions of compatibility, trust, and behavioral intention. While some relationships aligned with previous research, others yielded unexpected results, highlighting the complex dynamics of AI adoption in educational contexts. These findings not only contribute to the theoretical understanding of AI-supported OER development

and adoption but also offer practical implications for educators, policymakers, and technology developers seeking to enhance AI integration in teacher education.

One of the most unexpected findings of the study is that Compatibility → Trust hypothesis and Compatibility → Behavioral Intention was supported, while Trust → Behavioral Intention hypothesis was rejected. The theoretical implications of these findings highlight the critical role of compatibility in shaping both trust and behavioral intention in AI-assisted OER adoption among pre-service teachers. This suggests that when AI tools align well with educators' teaching practices and expectations, they are more likely to be trusted and adopted. However, the rejection of the Trust → Behavioral Intention hypothesis indicates that in the context of AI-assisted OER, trust alone may not be sufficient to drive behavioral intention; rather, compatibility and perceived usefulness may have stronger influences. Future research should further investigate the conditions under which trust becomes a significant factor in AI adoption, particularly in education.

From a practical standpoint, these results suggest that efforts to promote AI-assisted OER adoption among pre-service teachers should prioritize ensuring that AI tools seamlessly integrate with existing teaching practices and pedagogical needs. Training programs should focus on demonstrating how AI can enhance educators' workflows and instructional strategies rather than solely emphasizing its reliability. Additionally, developers and policymakers should consider designing AI-powered OER platforms that emphasize user-centered design, customization, and ease of integration to enhance compatibility perceptions. Since trust alone does not appear to directly influence adoption, institutional and professional support structures should focus more on showcasing AI's practical benefits rather than merely reassuring users of its credibility.

In case of hypotheses regarding Information Seeking, results emphasize the role of information-seeking behavior in shaping compatibility perceptions and behavioral intention in the adoption of AI-assisted OER. This aligns with the idea that pre-service teachers who actively seek information about AI tools are more likely to perceive them as suitable for their teaching practices and express a stronger intention to adopt them. However, the rejection of the Information Seeking → Trust hypothesis suggests that merely gathering information about AI does not necessarily translate into higher trust levels. This challenges conventional perspectives that assume greater knowledge leads to increased trust and indicates that trust in AI-assisted OER may be influenced by other factors, such as prior experiences, institutional endorsements, or perceived risks. Future research should explore the mediating or moderating variables that might explain this unexpected finding, such as transparency, explainability, or perceived ethical concerns of AI-powered tools.

Furthermore, these findings highlight the importance of fostering information-seeking behaviors among pre-service teachers to improve their perceptions of AI-assisted OER's compatibility and encourage adoption. Teacher education programs should incorporate hands-on experiences and structured exploration of AI tools to facilitate informed decision-making. However, since increased information-seeking does not inherently enhance trust, strategies beyond providing information—such as trust-building measures through transparent AI design, peer testimonials, and ethical

AI discussions—should be employed. Institutions and AI developers should also focus on improving AI explainability and user control to build trust rather than assuming that more exposure to information will automatically lead to greater confidence in AI-assisted OER.

The results provide important insights into the cognitive and affective factors influencing pre-service teachers' adoption of AI-assisted OER. The Task Efficiency → Compatibility relationship suggests that when AI tools streamline the OER development process and enhance productivity, pre-service teachers perceive them as well-suited to their teaching tasks. This supports technology acceptance models that emphasize task-technology fit as a determinant of adoption. However, the rejection of the Task Efficiency → Behavioral Intention hypothesis indicates that efficiency alone does not directly drive adoption. This suggests that while AI tools may optimize workflows, additional motivational or contextual factors (e.g., pedagogical value, institutional policies, or prior experiences) might play a stronger role in influencing actual adoption intentions.

Similarly, the findings related to Enjoyment highlight the importance of affective factors. The support for Enjoyment → Compatibility and Enjoyment → Behavioral Intention aligns with existing research suggesting that positive emotional experiences with technology foster greater perceived suitability and willingness to use it. However, the rejection of the Enjoyment → Trust hypothesis suggests that while pre-service teachers may find AI-assisted OER development engaging and enjoyable, enjoyment alone does not necessarily translate into trust. Trust may be shaped more by factors such as perceived reliability, ethical concerns, or institutional endorsement rather than enjoyment-driven experiences.

For teacher education programs and policymakers, these findings highlight the need for both efficiency and engagement in promoting AI-assisted OER adoption. Simply demonstrating the productivity benefits of AI is insufficient. Educators must also find the tools meaningful and aligned with their instructional needs. Training programs should emphasize not just how AI can save time, but also how it can be pedagogically valuable and adaptable to diverse teaching contexts. The enjoyment factor presents a strong case for designing AI-powered OER tools that are user-friendly and engaging. Developers should focus on intuitive interfaces, gamification elements, and interactive features to enhance user experience. However, since enjoyment does not directly lead to trust, additional strategies such as transparent AI processes, ethical AI discussions, and institutional validation should be incorporated to build confidence in AI-assisted OER development. Institutions should also provide clear guidelines on AI usage, addressing potential concerns related to bias, data privacy, and reliability to strengthen trust alongside enjoyment. Overall, these findings suggest that successful adoption of AI-assisted OER requires a balance between efficiency, enjoyment, and trust-building efforts, ensuring that pre-service teachers not only perceive AI tools as compatible and engaging but also feel confident in their reliability and ethical integrity.

The findings highlight the nuanced role of social factors in shaping pre-service teachers' adoption of AI-assisted OER. The support for Social Interaction → Trust suggests that AI tools fostering communication, collaboration, and knowledge-sharing contribute to increased trust in AI-assisted OER. This aligns with socio-cognitive

theories emphasizing that interpersonal interactions play a key role in trust formation, particularly in technology-mediated learning environments. However, the rejection of Social Interaction → Behavioral Intention implies that while social interactions help build trust, they do not directly translate into stronger adoption intentions. This suggests that pre-service teachers may require additional motivational drivers, such as perceived usefulness, ease of use, or institutional endorsement, before committing to AI-assisted OER.

The rejection of Social Influence → Compatibility suggests that despite external encouragement or peer recommendations, pre-service teachers may not necessarily view AI-assisted OER as aligned with their teaching practices unless they personally perceive its pedagogical value. However, the acceptance of Social Influence → Behavioral Intention indicates that social norms and the opinions of influential figures (e.g., faculty members, peers) still play a role in shaping pre-service teachers' willingness to adopt AI tools, even if these opinions do not alter their perceived compatibility. Finally, the rejection of Social Influence → Trust suggests that trust in AI-assisted OER is not simply built through external influence but rather through direct experience, perceived transparency, and the reliability of the technology itself.

From a practical standpoint, these findings suggest that fostering collaborative learning environments in teacher education programs can enhance trust in AI-assisted OER. Institutions should encourage group-based projects, peer discussions, and knowledge-sharing activities using AI tools to strengthen trust-building mechanisms. However, since trust alone does not directly lead to adoption, additional strategies—such as demonstrating the practical benefits of AI for teaching and learning—should be implemented to translate trust into behavioral intention. The findings also emphasize the importance of peer and faculty influence in promoting adoption. While social influence does not enhance perceived compatibility, it does affect behavioral intention. This suggests that institutional strategies should focus on role modeling and endorsement—for example, experienced educators demonstrating AI-assisted OER in practice, success stories from peers, and mentorship programs guiding pre-service teachers in AI integration. Furthermore, since social influence does not directly build trust, institutions must provide clear guidelines on AI reliability, ethical considerations, and transparency to foster confidence in AI-assisted OER beyond social endorsement. Overall, these results suggest that while social interactions and influence play key roles in shaping AI-assisted OER adoption, they must be complemented with pedagogical alignment, institutional support, and trust-building measures to maximize their impact on pre-service teachers' adoption behaviors.

Limitations and future directions

This study provides valuable insights into the adoption of AI-assisted open educational resources (OER) among pre-service teachers; however, several limitations should be acknowledged. First, the study was conducted with a specific group of pre-service teachers enrolled in an "Open and Distance Learning" course at a single

institution. While the sample includes participants from multiple departments, the findings may not fully generalize to pre-service teachers in different educational contexts, institutions, or cultural settings. Future research should replicate this study across diverse teacher education programs and geographical regions to enhance generalizability. Second, the study employed a cross-sectional research design, capturing participants' perceptions at a single point in time. Given that technology adoption is often a dynamic process influenced by evolving experiences and external factors, longitudinal studies are needed to track changes in pre-service teachers' attitudes, trust, and behavioral intentions over time. Such studies could provide deeper insights into how continued exposure to AI-assisted OER influences adoption behaviors and whether initial concerns (e.g., trust issues) diminish with prolonged use.

Third, while the study examined key constructs from the literature (Technology Acceptance Model, Uses and Gratifications Theory) and related theories, other potential factors influencing adoption were not explored. Variables such as perceived risk, ethical concerns, institutional support, and self-efficacy in AI use could significantly impact pre-service teachers' trust and behavioral intentions. Future research should expand the model to incorporate these additional factors to provide a more comprehensive understanding of AI-assisted OER adoption. Finally, this study relied on self-reported survey data, which may be subject to social desirability bias or limited by participants' subjective perceptions. Incorporating qualitative methods, such as interviews or focus groups, could help uncover deeper insights into the reasons behind specific adoption behaviors and trust dynamics. Additionally, experimental or observational studies that track actual usage behaviors of AI-assisted OER could complement self-reported data and provide a more objective measure of adoption trends.

Despite these limitations, this study offers a solid foundation for understanding the factors influencing pre-service teachers' adoption of AI-assisted OER. Addressing these limitations in future research will further refine theoretical models and inform educational policies and practices for integrating AI tools in teacher training programs.

Ethical approval

Prior to data collection, the study was approved by the institutional review board of Van Yuzuncu Yil University (Approval number: 2025/04-31).

Disclosure statement

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Data availability statement

The data of this study are available from the corresponding author upon reasonable request.

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